Rising Waters: A Machine Learning Approach to Flood Prediction

Project Initialization and Planning Phase

Project Proposal (Proposed Solution) report

The proposal report aims to transform loan approval using machine learning, boosting efficiency and accuracy. It tackles system inefficiencies, promising better operations, reduced risks, and happier customers. Key features include a machine learning-based credit model and real-time decision-making.

Project Overview	
Objective	The primary objective is to revolutionize the loan approval process by implementing advanced machine learning techniques, ensuring faster and more accurate assessments.
Scope	The project comprehensively assesses and enhances the loan approval process, incorporating machine learning for a more robust and efficient system.
Problem Statement	
Description	Addressing inaccuracies and inefficiencies in the current loan approval system adversely affects operational efficiency and customer satisfaction.
Impact	Solving these issues will result in improved operational efficiency, reduced risks, and an overall enhancement in the lending process, contributing to customer satisfaction and organizational success.
Proposed Solution	
Approach	Employing machine learning techniques to analyze and predict creditworthiness, creating a dynamic and adaptable loan approval system.
Key Features	- Implementation of a machine learning-based credit assessment model.

	- Real-time decision-making for quicker loan approvals. - Continuous learning to adapt to evolving financial landscapes.

Resource Requirements

Resource Type	**Description**	**Specification/Allocation**
Hardware		
Computing Resources	Processor requirements for model training and application execution	Intel i3 processor or above
Memory	RAM required for data processing and model execution	Minimum 4 GB RAM (8 GB recommended)
Storage	Storage space required for dataset, trained models, and application files	Minimum 2 GB free storage
Internet Connection	Required for dataset download, library installation, and cloud deployment	Stable internet connection
Software		
Operating System	Platform for project development and execution	Windows / Linux / macOS
Programming Language	Used for machine learning and application development	Python 3.8 or above
Frameworks	Web application development framework	Flask Framework
Libraries	Required libraries for data processing, visualization, and machine learning	NumPy, Pandas, Scikit-learn, XGBoost, Matplotlib, Seaborn, Joblib
Development Environment	Tools used for coding and model development	Anaconda Navigator, Jupyter Notebook, Visual Studio Code / PyCharm
Data		
Dataset Source	Historical weather and flood-related dataset	Open-source dataset from Kaggle
Data Format	Format used for storing and processing data	CSV format
Dataset Features	Weather parameters used for prediction	Annual rainfall, seasonal rainfall, cloud visibility, meteorological parameters

Machine Learning Resources		
Algorithms	Classification algorithms used for prediction	Decision Tree, Random Forest, KNN, XGBoost
Model Storage	Storage of trained machine learning model	floods.save
Preprocessing Storage	Storage of feature scaling model	scaler.save
Deployment		
Cloud Platform	Platform for scalable deployment	IBM Cloud
Web Server	Server used for hosting application	Flask Development Server